עיבוד שפה טבעית – תרגיל 1

1) א) בסה"כ ללא התחשבות בסוג האות, ישנן 9 תבניות המתחילות ברצף התווים 'ea' ו"types":

```
guykomash@guy:~$ tr -sc 'A-Za-z' '\n' < wizard.txt | sort | uniq | grep -i '\bea'
Each
East
each
ear
earth
easy
eat
eating
eats
guykomash@guy:~$ tr -sc 'A-Za-z' '\n' < wizard.txt | sort | uniq | grep -ic '\bea'
9
```

ב) יש 26 תבניות "tokens" בקורפוס המתחילות ברצף 'ea':

```
guykomash@guy:~$ tr -sc 'A-Za-z' '\n' < wizard.txt | grep -i '\bea'
easy
each
eats
eating
East
East
East
East
East
East
East
East
eat
eat
each
each
eat
earth
each
Each
Each
each
easy
earth
ear
easy
guykomash@guy:~$ tr -sc 'A-Za-z' '\n' < wizard.txt | grep -ic '\bea'
26
```

$$F_1 = \frac{(1^2+1)PR}{1^2P+R} = \frac{2PR}{P+R} = \left(1\right) \quad (2.1$$

$$\frac{2}{\frac{P+R}{PR}} = \frac{2}{\frac{1}{R}+\frac{1}{P}} \Rightarrow \text{הרמוני ממוצע}$$

ממוצע הרמוני קטן או שווה לממוצע חשבוני, ולכן:

$$F_1 = \underset{\text{הרמוני}}{\text{ממוצע}} \leq \underset{\text{חשבוני}}{\text{ממוצע}} = \frac{R+P}{2}$$

.

2) הטענה אינה נכונה, לפחות עם $P=0.2,\ R=0.8,\ \beta=5$

$$F_\beta(P,R) = \frac{(\beta^2+1)PR}{\beta^2P+R}$$

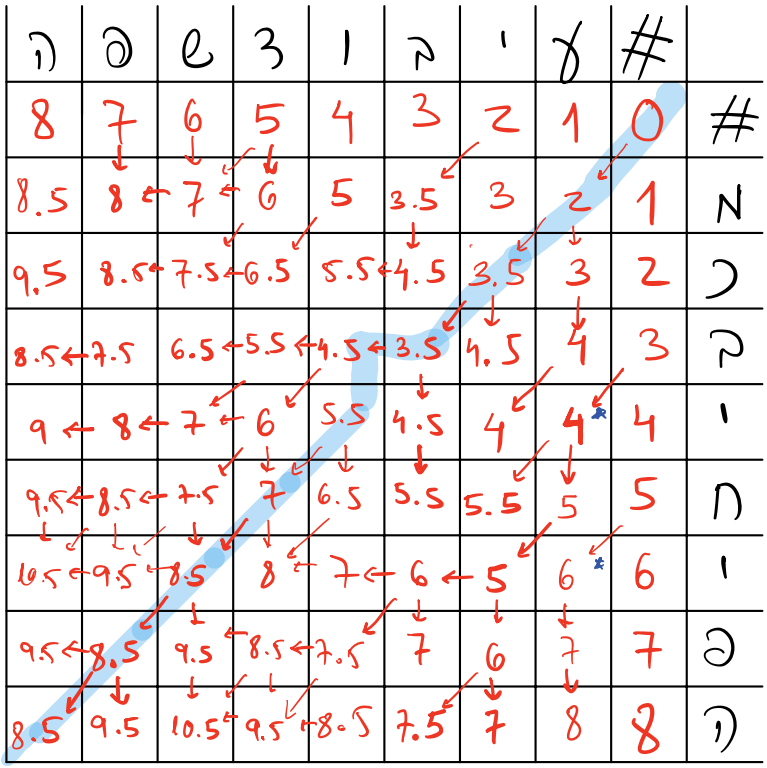
$$F_5(0.2,0.8) = \frac{(5^2+1)\cdot 0.2\cdot 0.8}{5^2\cdot 0.2+0.8} = \frac{104}{145} \approx 0.717$$

$$\underset{\text{חשבוני}}{\text{ממוצע}} = \frac{P+R}{2} = \frac{0.2+0.8}{2} = \frac{1}{2}.$$

$$F_5(0.2,0.8) \approx 0.717 \neq \frac{1}{2}$$

הפרכנו את הטענה עבור $P=0.2,\ R=0.8,\ \beta=5$

(2.2


ה ם ש ן ו ר י ץ # _
8 7 6 5 4 3 2 1 0 #
8.5 8 7 6 5 3.5 3 2 1 ן
9.5 8.5 7.5 6.5 5.5 4.5 3.5 3 2 ך
8.5 7.5 6.5 5.5 4.5 3.5 4.5 4 3 ר
9 8 7 6 5.5 4.5 4 4 4 י
9.5 8.5 7.5 7 6.5 5.5 5.5 5 5 ן
10.5 9.5 8.5 8 7 6 5 6 6 י
9.5 8.5 9.5 8.5 7.5 7 6 7 7 ם
8.5 9.5 10.5 9.5 8.5 7.5 7 8 8 ן

(1 (3

$$f_1 = (1, 1, 1, 1),\ y = 1$$
$$f_2 = (1, 0, 3, 0),\ y = 0$$
$$f_3 = (0, 1, 0, 1),\ y = 1$$

(2

$$\theta^{(0)} = \langle 0, 0, 0, 0, 0 \rangle,\ b^{(0)} = 0$$

$$\eta^{(t)} = \frac{0.8}{t},\quad \eta^{(1)} = \frac{0.8}{1}$$

מעבר ראשון

$$\hat{y} = \sigma(\theta^{(0)} \cdot f_1 + b) = \sigma(0) = \frac{1}{2}$$

$$\theta^{(1)} = \theta^{(0)} - \eta^{(1)} \left(\nabla L_{ce}(\theta^{(0)}) \right)$$

$$\theta^{(1)} = \langle 0, 0, 0, 0, 0 \rangle - \frac{0.8}{1} \langle -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2} \rangle$$

$$= \langle \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5} \rangle$$

$f_2 = \langle 1, 0, 3, 0 \rangle \quad , y = 0$

דוגמה מספרית

$\hat{y} = \sigma(\theta^{(1)} \cdot f_2 + b) = \sigma\left(\frac{2}{5} + 0 + \frac{6}{5} + 0 + \frac{2}{5}\right)$

$= \sigma(2) \approx 0.88 \Rightarrow \hat{y} - y \approx 0.88$

$\theta^{(2)} = \theta^{(1)} - \eta^{(2)} \left(\nabla L_{ce}\, \theta^{(1)}\right) (\hat{y} - y) f_2^{(1)}$

$= \left\langle \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5} \right\rangle - \frac{0.8}{2} \langle 0.88, 0, \frac{66}{25}, 0, 0.88 \rangle$

$\left\langle \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5}, \frac{2}{5} \right\rangle - \left\langle \frac{44}{125}, 0, \frac{132}{125}, 0, \frac{44}{125} \right\rangle$

$= \left\langle \frac{6}{125}, \frac{2}{5}, \frac{-82}{125}, \frac{2}{5}, \frac{6}{125} \right\rangle$

דוגמה מספרית

↙ bias

$f_3 = \langle 0, 1, 0, 1, 1 \rangle \quad y = 1$

$\hat{y} = \sigma(\theta^{(2)} \cdot f_3) = \sigma\left(0 + \frac{2}{5} + 0 + \frac{2}{5} + \frac{6}{125}\right) =$

$\sigma\left(\frac{106}{125}\right) \approx 0.7 \Rightarrow \hat{y} - y \approx -0.3$

$\theta^{(3)} = \theta^{(2)} - \eta^{(3)} \left(\nabla L_{ce}(\theta^{(2)})\right) =$

$\left\langle \frac{6}{125}, \frac{2}{5}, \frac{-82}{125}, \frac{2}{5}, \frac{6}{125} \right\rangle - \frac{0.8}{3} \left\langle 0, -\frac{3}{10}, 0, -\frac{3}{10}, -\frac{3}{10} \right\rangle$

$$= \left\langle \frac{6}{125}, \frac{2}{5}, \frac{-82}{125}, \frac{2}{5}, \frac{6}{125} \right\rangle + \left\langle 0, \frac{2}{25}, 0, \frac{2}{25}, \frac{2}{25} \right\rangle$$

$$\left\langle \frac{6}{125}, \frac{12}{25}, -\frac{82}{125}, \frac{12}{25}, \frac{16}{125} \right\rangle \swarrow$$

$$f_4 = \langle 0, 0, 1, 1, 1 \rangle \qquad (3$$

$$\hat{y} = \sigma(\theta \cdot f_4) = \sigma\left(0 + 0 - \frac{82}{125} + \frac{12}{125} + \frac{16}{125}\right) = \sigma\left(-\frac{54}{125}\right)$$

$$\approx 0.39$$